Rig Rust Repair: a quadruped that finds and fixes pipe corrosion

Unitree Go2 + 0–0.45 m lift mast + UR3 arm on an offshore oil rig in Isaac Sim 6.2. It detects rust, sprays, soaks 30 s, scrubs with a PPO policy, verifies and paints, and every step is decided by what its camera sees.

Agentverse 2026 · Robotics track
Technical summary

USER AND PROBLEM

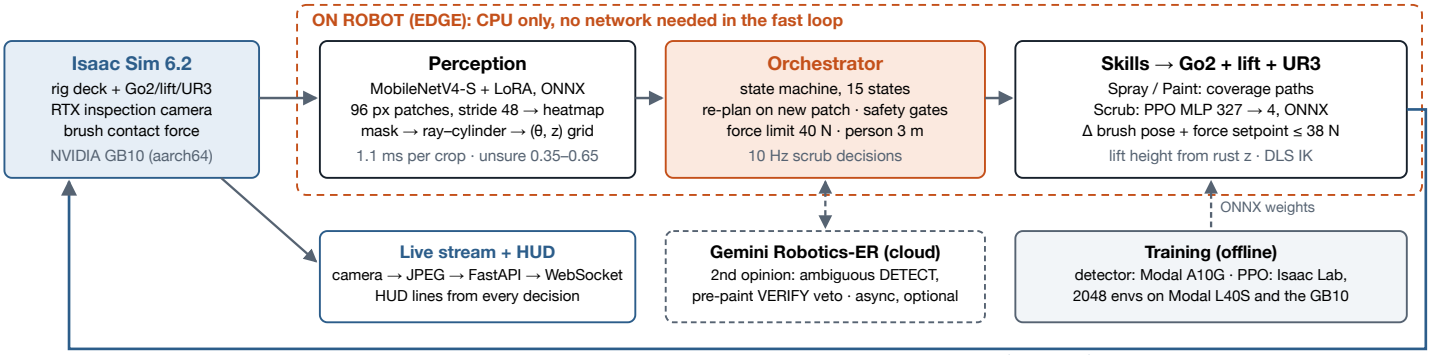
User: asset-integrity and maintenance crews on offshore platforms. Topside pipework corrodes all the time in salt spray. Today people find it from rope access or scaffolding near live hydrocarbons, then fix it by hand with remover, wire brush and paint.

Why it matters: the work is **dangerous** (height, weather, possible gas release), **costly** (every task needs a person flown offshore) and **never finished** (kilometres of pipe per platform keep corroding). A robot that patrols, judges and does the routine repair, and hands only the doubtful cases to a person, keeps people out of the hazard zone.

WHAT IT DOES

- **Perception decides the action.** The rust heatmap is projected onto a (θ, z) grid on the pipe; that grid sets the spray path, scrub targets, paint path and when the repair is done.
- **Adapts mid-run.** New rust during the soak or scrub → re-plan: spray the new patch, extend the soak. Weld bead → back off. Glare → second view. Lens mist → hold. Worker within 3 m → pause.
- **Grounded scrubbing.** Archard-style wear (removal ~ force × speed), 4× faster on sprayed and soaked cells, so the chemistry step matters to the policy.
- **Auditable.** Every transition logs detector confidence, rust fraction, patch ids and force to `decisions.jsonl` and the HUD.

SYSTEM ARCHITECTURE



PATROL → **DETECT** ⇄ SECOND_VIEW → **APPROACH** → **SPRAY** → **SOAK 30 s** (new patch ⇒ re-spray) → **SCRUB (PPO)** ⇄ BACK_OFF → **VERIFY** (>5 % rust ⇒ scrub, ≤3 cycles) → **PAINT** → **VERIFY_PAINT** → **DONE** · any state → HOLD / PAUSE_PERSON / ESCALATE_TO_HUMAN

WHERE EACH MODEL RUNS

Component	Runs on	Latency / rate
Rust classifier (2.49 M, 0.19 GMACs)	robot CPU, ONNX	1.1 ms (4 thr) · 2.6 ms (1 thr)
Localizer (patches on 640×360)	robot CPU	batched · every 5 s in soak
Scrub policy (MLP, 327 → 4)	robot CPU, ONNX	0.012 ms* · 10 Hz loop
Orchestrator + skills	robot CPU, Python	10 Hz scrub decisions
Gemini Robotics-ER 2 preview	cloud API, async	**TBD** · 20 s timeout
Isaac Sim 6.2 render + stream	GB10, 121 GB unified	RTX · JPEG over WebSocket
Detector training	Modal A10G	minutes per run
PPO training, Isaac Lab 2.3.2	Modal L40S / GB10	~45k / ~15k env-steps/s

*Apple M5 Max CPU, batch 1. Detector latency from the bake-off. GB10: 150 PPO iterations × 2048 envs in 524 s. Modal cannot render (gVisor, no Vulkan), so all RTX rendering runs on the GB10.

KEY RESULTS

Detector: MobileNetV4-S + LoRA r=16 (0.26 M trainable)	
Real-photo test (881): F1 / recall / precision / AUROC	**0.977** / 0.983 / 0.970 / 0.998
Held-out Isaac render crops (v1 → v2 model)	10/13 → **13/13**
Synthetic sim-domain test F1 (v1 → v2)	0.795 → 1.000

Scrub PPO · rig geometry · new patch each episode	**v1**	**v2**
Success (<5 % rust left) · episodes	100 % · 513	97.6 % · 1026
Episodes with force > 40 N	76.8 %	**1.0 %**
Mean / p95 episode-max force	49.9 / 70.1 N	**29.9 / 35.7 N**
Mean time to clean	5.2 s	6.8 s

v2 gives up 2.4 points of success to stay inside the force budget (fine-tuned with a quadratic penalty above 30 N, a 38 N setpoint cap and wider randomisation). Full repair loop in Isaac 6.2 with the trained models: **TBD**.

CRITICAL FAILURE MODES → RESPONSE (ON THE HUD)

Over-force	Force > 40 N → **BACK_OFF**: retract 5 cm, press × 0.7; 3 back-offs → **ESCALATE**. The policy is trained to stay under 40 N, and its setpoint is capped at 38 N outside the network.
Missed / residual rust	Painting over rust is the costliest error. The threshold is set for recall ≥ 0.95 in both domains. VERIFY needs <5 % rust, ≥80 % of the target visible and ≤5 % uncertain cells. Gemini can veto paint. Still rusty after 3 cycles → **ESCALATE_TO_HUMAN** with the residual cells.
Ambiguous view	p in 0.35–0.65 → up to 2 **SECOND_VIEW**s, then Gemini, then a human.
Occlusion	No frame or a flat frame → **HOLD**; escalate after 5 s.
Person nearby	Worker < 3 m → **PAUSE_PERSON**; resume when clear.

SAFETY DESIGN

- Conservative by default: in doubt it looks again, holds or asks a person, and it never paints when unsure.
- Hard limits live outside the learned policy: the 40 N gate, the 38 N clamp, the 3 m person radius, and caps on cycles, back-offs and holds.
- The fast loop never waits on the cloud. With no key, network or valid reply, it runs exactly as it does without Gemini.
- Escalations carry evidence (frame, confidence, residual cells), so a person can decide quickly.

LIMITS

- Simulation only. The bridge poses the robot kinematically (FK/IK, not PhysX), the legs are fixed while scrubbing, and the brush force is an analytic spring, not a contact sensor.
- The scrub policy trains on privileged grid state; in deployment, the localizer fills the grid.
- There are only 13 sim-render test crops and no real painted-steel negatives. One real source has up to 13 % false positives on weathered surfaces.
- The rusty deck grate reads as rust, so only pipe pixels are classified; the rest are masked grey.
- The arm links do not collide with the pipe, and the IK does not hold the brush normal to the surface.

Sources: `detection/RESULTS.md`, `policy/SCRUB.md`, `runs/scrub_v2/eval_rig_meta_forcepatch.json`, `runs/scrub_watson/eval_rig_meta_forcepatch.json`, `integration/INTERFACE.md`, `integration/SIM_BRIDGE.md`, `docs/diagrams.md`. Live stream: rig.zeecloud.org. TBD = still being produced.